HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Managed end-to-end technology transfer for new Active Pharmaceutical Ingredient (API) product introductions within R&D;, ensuring seamless scale-up from lab to commercial manufacture and **process robustness**.
- Led cross-functional teams of process chemists and engineers to execute manufacturing campaigns, ensuring **cGMP compliance** and repeatable product quality for multiple concurrent projects in a fast-paced CDMO environment.
- Developed and executed detailed technology transfer plans, interpreting **process data**, analytical results, and development reports to identify and mitigate scale-up risks.
- Collaborated with external customers and stakeholders to gather requirements, define project scopes, and communicate technical progress, ensuring alignment with customer expectations.
- Performed root cause analysis and problem-solving for process deviations, implementing continuous improvement initiatives to enhance manufacturing efficiency and product quality.
- Ensured adherence to stringent **regulatory and safety guidelines** throughout the technology transfer lifecycle, maintaining comprehensive audit trails and documentation.
- Containerized key process simulation tools using **Docker**, facilitating consistent deployment and execution across development and testing environments on **AWS EC2**.

SKILLS

Process Engineering – Technology Transfer, API Manufacturing, Scale-up, Reaction Kinetics, Mass Transfer, Crystallization, Isolation, Drying, Process Data Interpretation, cGMP Compliance

Project Management – Cross-functional Team Leadership, Project Planning, Stakeholder Communication, Risk Mitigation, Root Cause Analysis, Continuous Improvement

Software & Tools – Dynochem (Desirable), Computational Fluid Dynamics (Desirable), Splunk, Dynatrace, Jira, Confluence

Cloud & Containerization – AWS (EC2, S3, Lambda, CloudWatch), Docker

Programming & Scripting – Python, Java, SQL

PROJECTS

API Process Scale-Up and Validation (HCL Technologies)

- Engineered a Python-based data analysis pipeline to process and visualize experimental data from pilot-scale API synthesis runs, identifying critical parameters impacting yield and purity.
- Developed automated reporting scripts using **Python** and **SQL** to track key performance indicators for process validation batches, reducing manual reporting effort by approximately 40% for the operations team.
- Collaborated with senior engineers to define integration requirements for a new batch execution system, ensuring seamless data flow and audit trail integrity for regulatory compliance.

Technology Transfer Process Simulation and Optimization (HCL Technologies)

- Built and maintained **Java Spring Boot** microservices responsible for ingesting and transforming laboratory process data into a format compatible with plant-scale simulation software.
- Implemented RESTful APIs for data exchange between the simulation environment and the enterprise data warehouse, supporting approximately 500 requests per hour during peak analysis periods.
- Contributed to the containerization of simulation modules using **Docker**, improving deployment consistency and reducing setup time for new transfer projects.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)